

Data Visualization and Exploration

COMS7056A/COMS4060A

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Lecture 5



Lesson Plan

- Time Series Data
 - Stationarity
 - Autocorrelation
 - Dicky Fuller test
 - Feature Extraction
 - Trend
 - Seasonality
 - Resampling – fixed vs irregular frequency
 - Windowing – overlapping and non-overlapping
 - Autoregression
 - FTs
 - Frequency spectrum
 - Smoothing (exponential and sma)
 - Power spectral density

Resources

- <https://arxiv.org/pdf/1610.07717.pdf>
- Doing Data Science, Cathy O'Neil & Rachel Schutt, pg 135-164
- Feature Engineering for Machine Learning and Data Analytics, G Dong & H Liu, pg 87-116
- Python for Data Analysis, Wes McKinney, pg 289-328
- <http://www.statslab.cam.ac.uk/~rrw1/timeseries/t.pdf>
- <http://www.db.ucsd.edu/static/TimeSeries.pdf>



Time Series data – an Intro

- Specifically, this refers to a series of data points that are time-ordered (and usually sequential)
- In time series data, there is often some correlation between adjacent data points, and we are typically looking to use this type of data to forecast future events
 - The key difference between a normal dataset used for an ML problem is that in those cases, historical data is weighted equally. In time series data, there should be a greater weight applied to the most recent observations
- Some of the models you might use for forecasting: autoregressive time-series models like (S)ARIMA, simple moving averages, LSTM neural nets, Transformers or linear feature-based models
 - Exploring the data by itself can give valuable insights too!
- Many of the models require the time series data to be stationary, this being one of the most important aspects of time series data exploration and preparation
- When exploring time series data, we want to look for stationarity, but also patterns or correlations that might indicate some kind of signal. The most obvious solution is to plot it, but generally we need more robust methods to investigate it. We'll look at:
 - Correlations between timestamps (autocorrelation)
 - Composition of the time series: trends (and cyclicity), seasonality, noise
 - Frequency based approaches
 - Feature engineering
- Based on what we find, we can choose suitable transformations and modelling techniques
- Some examples: weather data, accelerometer data, GPS, stock market data, sales reports, medical data like BPS etc
- Technical note: time series data can be viewed from the time or frequency domain. If you are familiar with Fourier Analysis (which is extremely powerful in this area), then you will be aware of the relationship between the two. Some of the techniques here focus on the frequency domain.



Time Series data

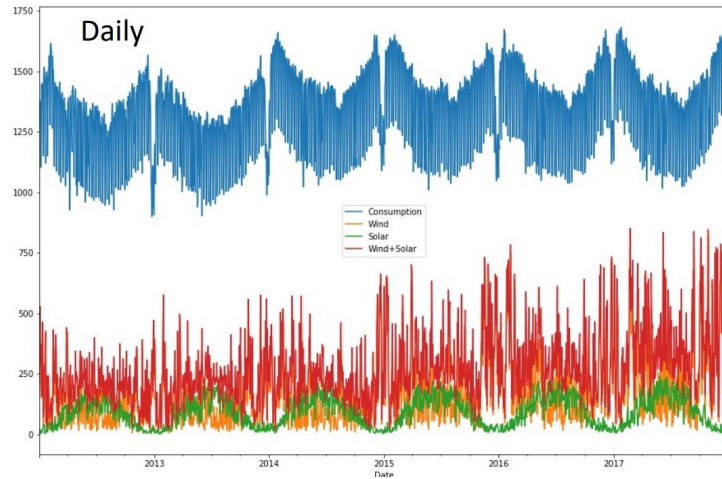
- What is the date format?
 - YYYY/MM/DD or DD/MM/YYYY or MM/DD/YYYY?!
- Is it in unix epoch?
 - Number of milliseconds since 1970
- What is the frequency? Is it fixed? Are there missing periods?
 - Can we patch missing values from another dataset?
- Timezones
 - Especially DST – DST has changed historically per country
 - Work in UTC as much as possible and express times as offsets
 - Convert to local tz only if needed
- Plotting – Make sure axes are correct
 - If plotting multiple datasets, make sure everything is on same scale
 - same frequency, same periods
- Data alignment issues – time series indexes don't match
- Missing values – how do we handle this??
 - Is it absolutely critical to have all readings? Is the data subject to random fluctuations or large fluctuations within the period of missing data? What would the adverse effects of this be?

What is the frequency?

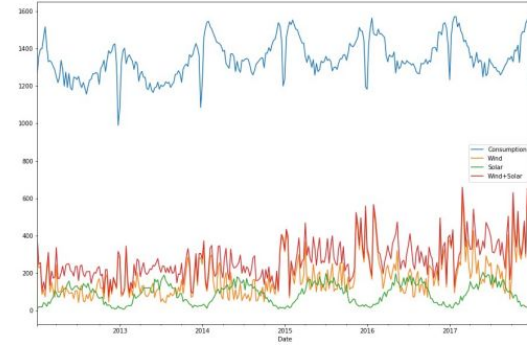
- What is the difference in time between the timestamps (ie period)?
 - Is it seconds, minutes, hours?
- Is the period useful for the intended purpose?
 - Should we be rolling it up to days, months, quarters (eg. Financial fiscal year end), years, etc.?
 - Should we be trying to break it into even smaller periods?
- Upsampling – moving to a higher frequency
 - Need to interpolate the data somehow
 - We can replace the missing entries using forward filling
- Downsampling – moving to a lower frequency
 - How to label this? Beginning or end of dataset? Which bin edge is inclusive?
 - Open High Low Close resampling for finance
 - Groupby and aggregate – mean, for example

When to resample?

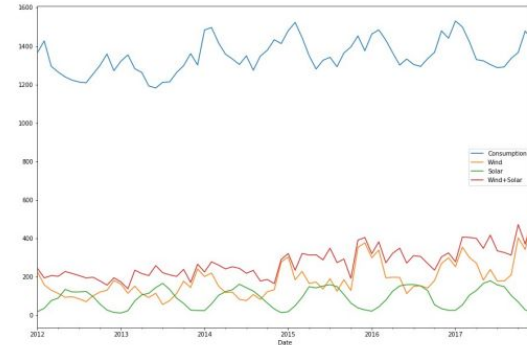
- This energy consumption data is really noisy – we could either downsample or smooth it, either will work



Weekly

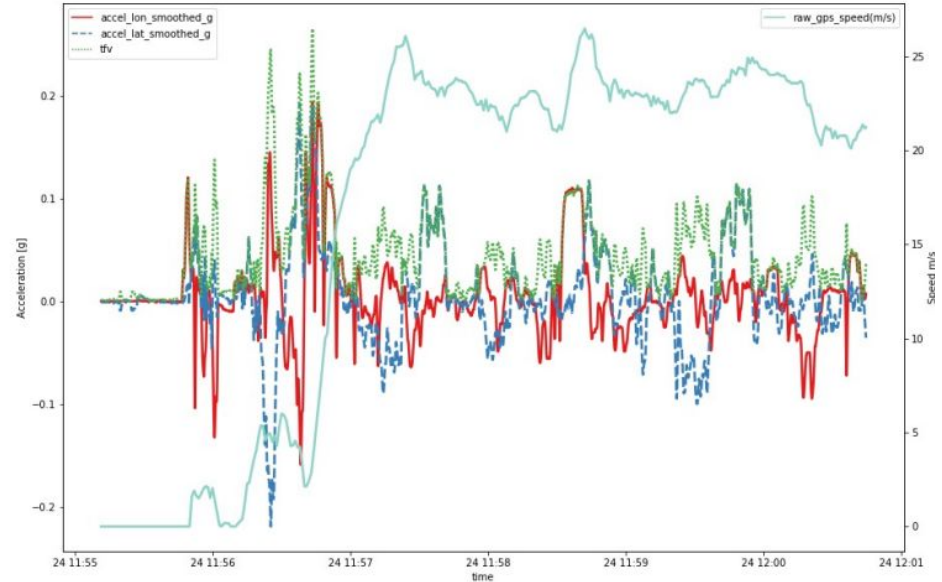


Monthly



Resample: acceleration data

- GPS data is sampled at 1Hz
- Accelerometer at 15Hz
- Here, there is upsampling done on the GPS data, using a moving average function
- Acceleration data could be downsampled, *but*, we in this case, we're more interested in the changes in acceleration, so we don't want to lose resolution



Smoothing: moving averages

- The moving averages are based on the idea of windowing:
 - A lookback window of length X is defined over which the moving averages are calculated
 - Window length should be greater than the frequency of the time series, and is typically defined as a minute, hour, day, week or month (as is applicable)
- Simple Moving Average: each next value is the mean (or sum or variance) value of the previous X observations
 - Has the effect of smoothing out the data
 - This works if there are clear time periods in the data and, visually, the distribution looks somewhat regular
- If the data is more complex, where there are no clear time periods, or the time periods are irregular, we can use an (exponentially) weighted moving average
 - Most recent values are weighted more than past values
 - An example would be stock prices, where the data isn't regular, and there is no obvious regularity
 - The simplest one would be something like
 - $y = \alpha x_t + (1 - \alpha)y_{t-1}$ where $t > 0$ and α is between 0 and 1. α is a smoothing factor that determines the weight decay
 - Again, no simple answer for what values of α to use!
 - An exponentially weighted average is similar to the weighted average, except that the α parameter is raised to a power
 - There are more complex ones like double and triple exponential averages, but these typically are used to suppress trends and seasonality too
- Important to note that because the windows are of length X , the first $X-1$ values of the moving average are undefined!

Missing data

- Any number of interpolations or imputations can be used to fill in missing data, or, if necessary, these entries can be dropped
- Some typical options include
 - linear interpolation
 - spline interpolation
 - Don't use these where there is strong seasonality
 - use prior value
 - use next value
 - Most typical for time series data
 - simple moving average
 - weighted moving average
 - exponential moving average
- If there is seasonality, look at
 - split seasonality or
 - Removing seasonality followed by imputation or interpolation, then adding seasonality back in (we'll look at time series decomposition in a few slides)

Stationarity

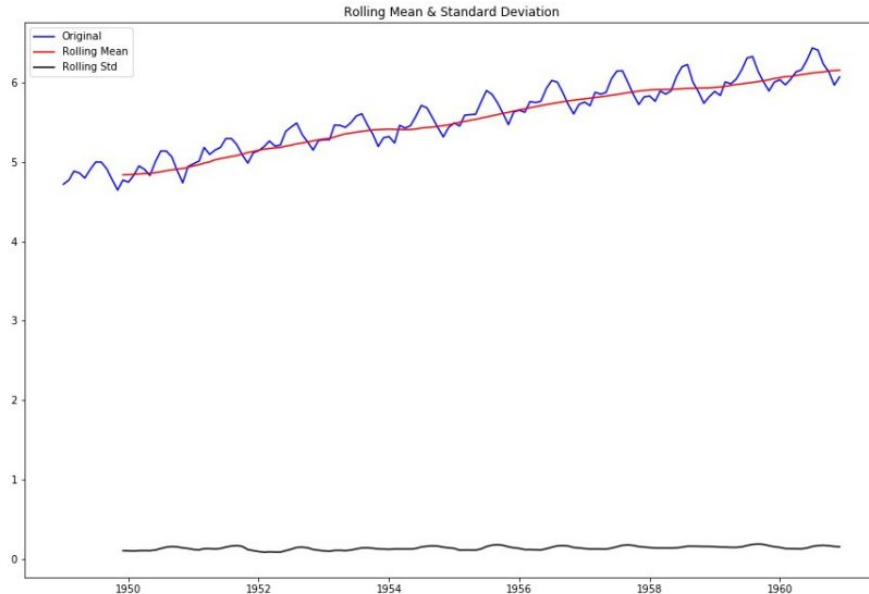
- Statistical properties and covariance independent of time
 - **mean and variance are constant**
 - The properties of the process *generating* the time series doesn't change
 - The time series can still change values over time, but the underlying mechanism does not change (consider a straight line with a positive gradient)
- Why do we care about stationarity?
 - stationary processes are easier to analyse
 - If the way in which a series changes is predictable, it's easier to make predictions
 - Circular reasoning here, but many of the existing tools for time series analysis assume a stationary time series
- How do we know if it is stationary?
 - Visual – can we see anything obvious?
 - Statistical test for a unit root
- Types of stationarity
 - Strict – mean, variance, covariance are not functions of time
 - Trend – No unit root, but there is still a trend in the data
 - Difference – made stationary by taking a difference (like ADF tests for)
- If it isn't stationary, we can *make* it stationary
 - We will look at some options in the next few slides
 - Introduce a difference (subtract from itself with a lag), smoothing, trend & seasonality removal (time-series decomposition), log transforms, box-cox transform

Stationarity tests

- A time series can be written as $y_t = \alpha \times y_{t-1} + \epsilon_t$ where y is the time series value at time step t , α is a scaling factor, and ϵ is the error term,
 - This means $y_{t-1} = \alpha \times y_{t-2} + \epsilon_{t-1}$
 - Giving $y_t = \alpha^n \times y_{t-n} + \sum \epsilon_{t-i} \alpha^i$
- If $\alpha = 1$, then $y_t = y_{t-n} + \sum \epsilon_t$, ie, the variance increases with time, meaning the series is not stationary
 - This is what is known as a unit root
- **Augmented Dickey-Fuller:**
 - Null hypothesis: that $\alpha=1$ and the process is not considered stationary
 - Alternative hypothesis: No unit root, so stationary
- It returns the following outputs:
 - The p-value
 - The value of the test statistic
 - Number of lags considered for the test
 - The critical value cut-offs
- **Reading the results:**
 - Test statistic must be lower than the critical value AND p-value must be < 0.05 implies stationary process
 - Otherwise, the null hypothesis cannot be rejected, and assumed non-stationary
- **KPSS (Kwiatkowski–Phillips–Schmidt–Shin) Test**
 - The null and alternate hypothesis for the KPSS test are opposite that of the ADF test
 - Null hypothesis: it is trend stationary
 - Alternative hypothesis: $\alpha=1$ and it has a unit root
 - Results are read the same way as for ADF

Dickey Fuller Test Example

Airline passenger data with log transform



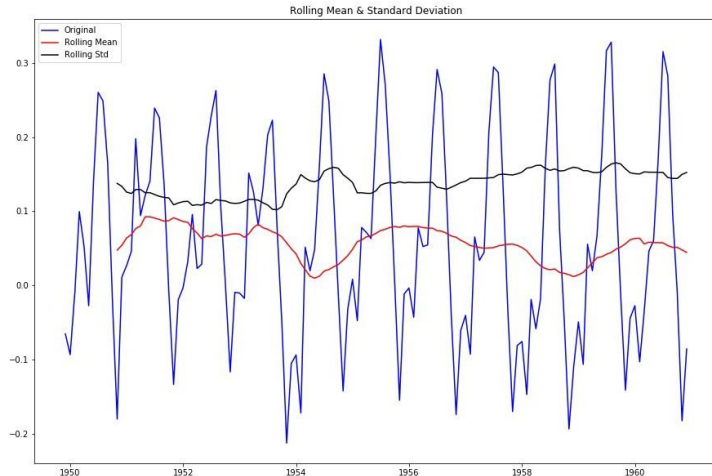
● Results of Dickey-Fuller Test:

- Test Statistic -1.717017 p-value 0.422367
- #Lags Used 13.000000
- Number of Observations Used 130.000000
- Critical Value (1%) -3.481682
- Critical Value (5%) -2.884042
- Critical Value (10%) -2.578770

Dickey Fuller Test Example

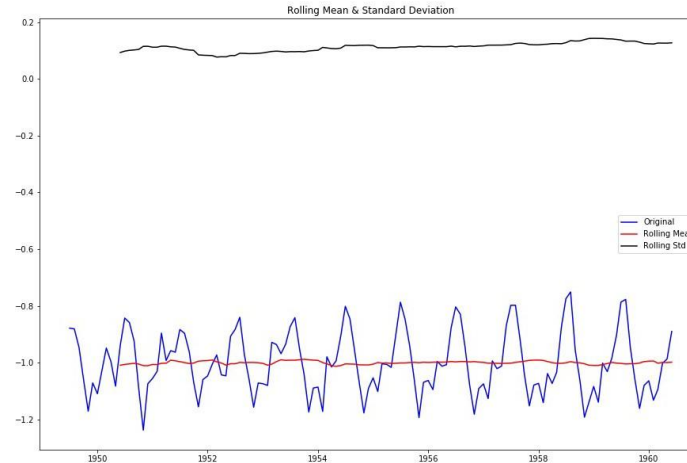
Airline passenger data with trend removed

- Results of Dickey-Fuller Test:
 - Test Statistic -3.162908
 - p-value 0.022235
 - #Lags Used 13.000000
 - Number of Observations Used 119.000000
 - Critical Value (1%) -3.486535
 - Critical Value (5%) -2.886151
 - Critical Value (10%) -2.579896



Airline passenger data with trend & seasonality removed

- Results of Dickey-Fuller Test:
 - Test Statistic -4.763795
 - p-value 0.000064
 - #Lags Used 13.000000
 - Number of Observations Used 118.000000
 - Critical Value (1%) -3.487022
 - Critical Value (5%) -2.886363
 - Critical Value (10%) -2.580009



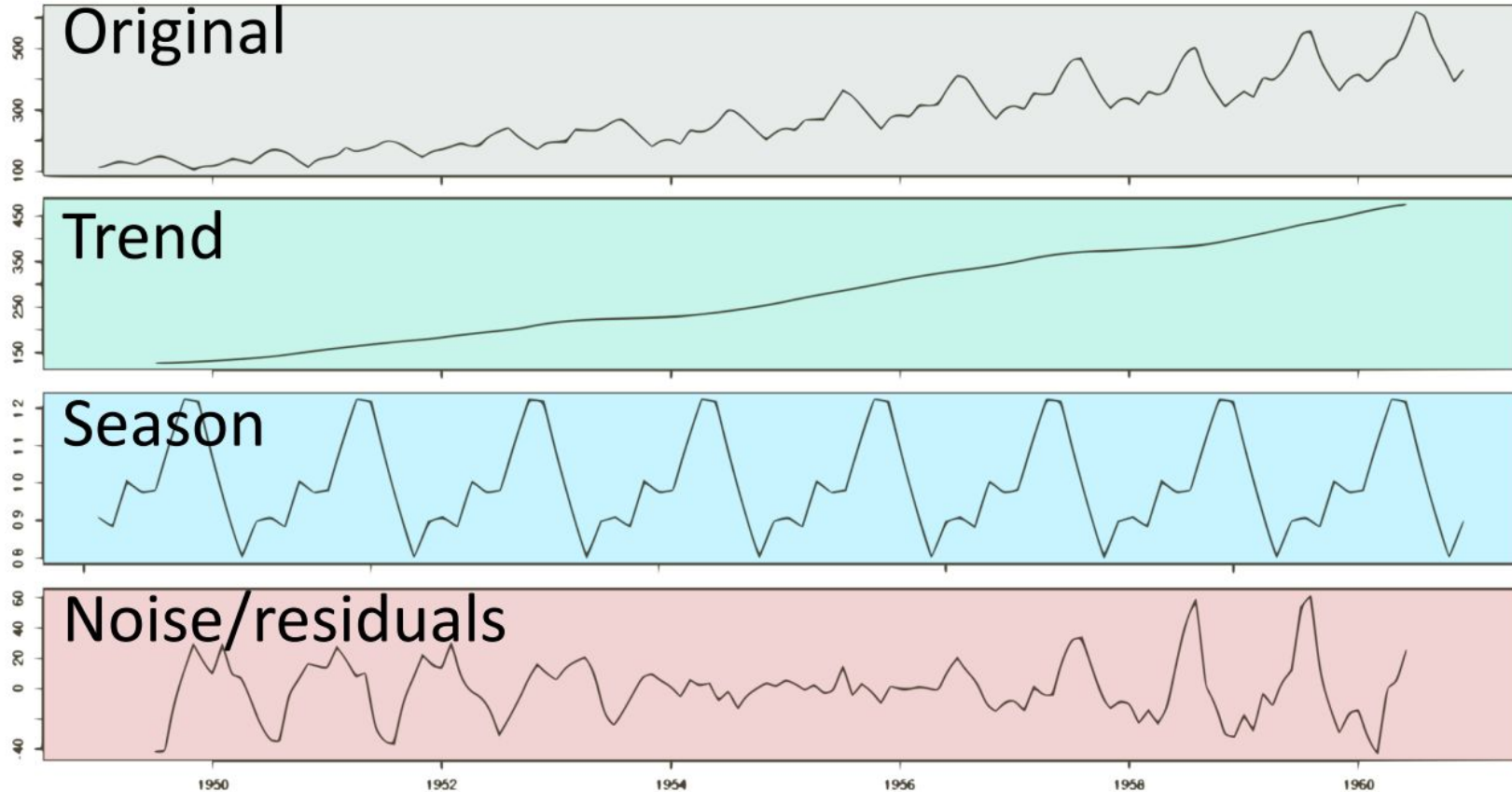
Towards Stationary

- There are a few general approaches we can take:
 - Transformations
 - Time series decomposition
 - Differencing (using lags)
- Common transformation methods include power transform, square root, and log transform
 - Example, the variance of the airline passenger plot is 14k, but if we take the log of this, it lowers to 0.195
 - Before taking the log, the test statistic for ADF is -3.22, with a p-value of 0.01, and after taking the log, it drops to -3.86 with a p-value of 0.002
- After these are performed, the new series is checked for stationarity, and once it is found, then the time series is ready for modeling!

Time series decomposition

- Basic components of a time series dataset (not necessarily all present):
 - Trend: is there an overall progression in the data (positive or negative), regardless of period. This is a low frequency component of the series
 - Seasonality: there are patterns to the data occurring in fixed periods (day, month, quarter, etc). Note that there can be multiple “seasons” in a dataset, making this aspect more complicated than it seems at first
 - Noise: parts of that data that can not be explained by trends or seasonality random, irregular terms
 - Cyclical: this is something that is cyclical, but not on a fixed timescale. For example, stock market crashes and boom periods. Normally, this gets included in the trend component
- We want to take a time series and break it up into these components, and model each one separately
- Additive model
 - Our overall time series is an additive combination of the trend, seasonality and noise
 - $y_i = t_i + s_i + n_i$, where t is trend, s is seasonality, n is noise, and i represents the i th timestep
- Multiplicative model
 - Our time series is a multiplicative combination of the trend, seasonality and noise
 - $y_i = t_i * s_i * n_i$, where t is trend, s is seasonality, n is noise and i represents the i th timestep
- We generally do not know a-priori if the composition will be additive or multiplicative, but we can make an informed guess:
 - If we plot the time series, and can see an obvious seasonality, we can assess this by looking at the proportion in the increase relative to the previous value
 - The trend might have to be removed first, either subtracting or dividing the the trend component, so that the seasonality is visible

Time series decomposition



Trends and seasonality, additive and multiplicative

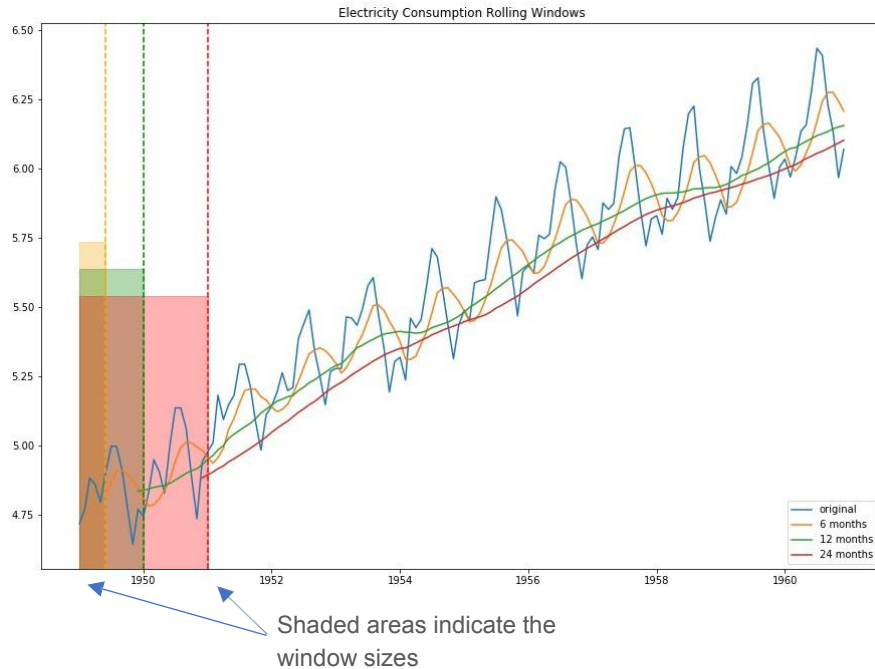
	Nonseasonal	Additive Seasonal	Multiplicative Seasonal
Constant Level (SIMPLE)	 NN	 NA	 NM
Linear Trend	 LN	 LA	 LM
Damped Trend (0.95)	 DN	 DA	 DM
Exponential Trend (1.05)	 EN	 EA	 EM

Identifying trends

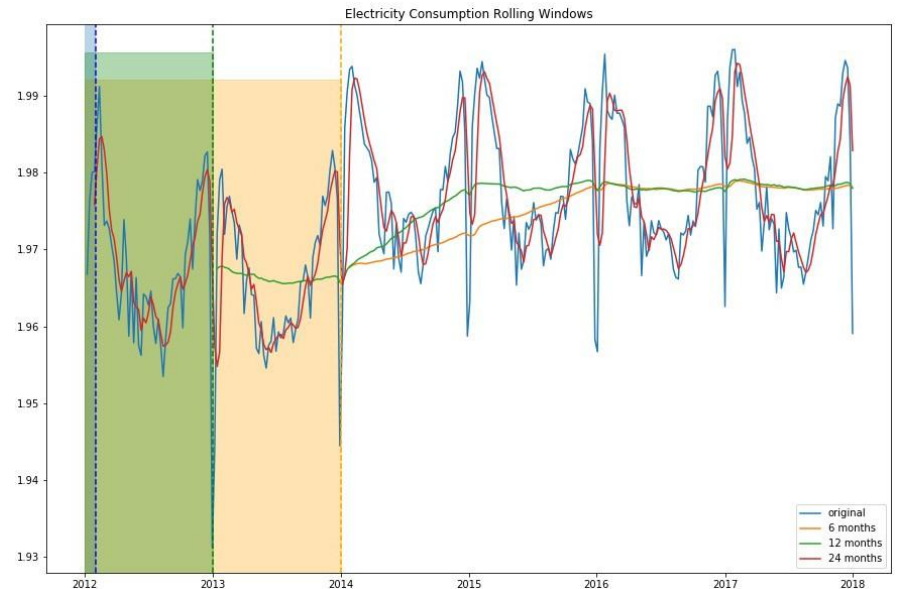
- A typical time series will have fluctuations that can obscure an underlying trend
 - If there is a trend, it's probably not a stationary process – we'll have to remove this!
- We want to remove these fluctuations and smooth out the data so that we can get an overall picture of what is happening
- If there are anomalous points, we can try a transform first: taking the log or square-root of the time series, or performing a Box-Cox transform (see first lecture), this is the first step towards smoothing the dataset
- Then, we can introduce a moving average:
 - Each next value is the mean value of the previous X observations (we say that we are looking at a window of size X)
 - There are a few different types of moving average we could use depending on the situation: sum, mean, weighted average, exponential moving average (we'll get to that shortly)
 - We can apply second and third order exponential moving averages to remove trend and seasonality at once
- Choosing the window size X is often an iterative approach: we want it to be about the length of any “season” in the data, but we don't always know what this is!
 - Choosing smaller windows will still smooth out the data, but not to the same extent
 - Choosing larger windows will result in a loss of too much information

Identifying trends

Airline passenger moving averages



Electricity consumption moving averages

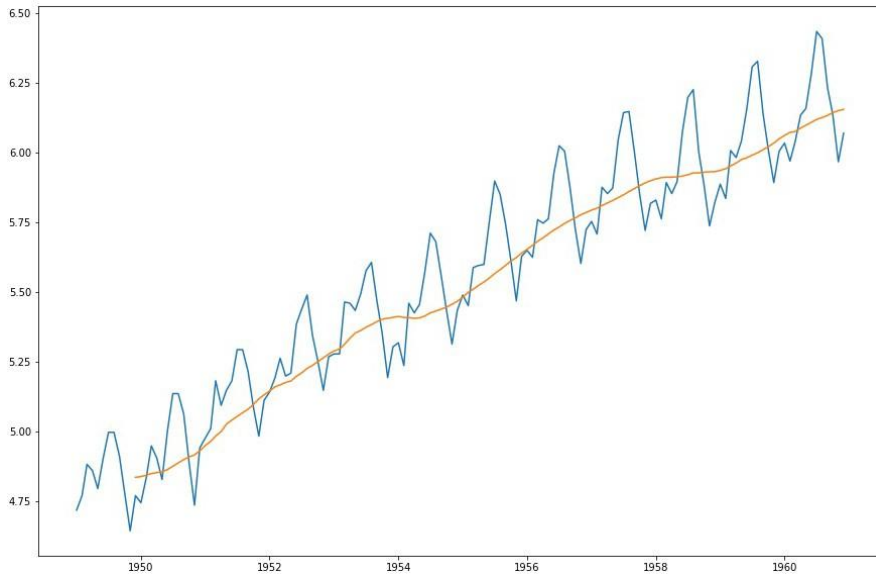


Removing the trend

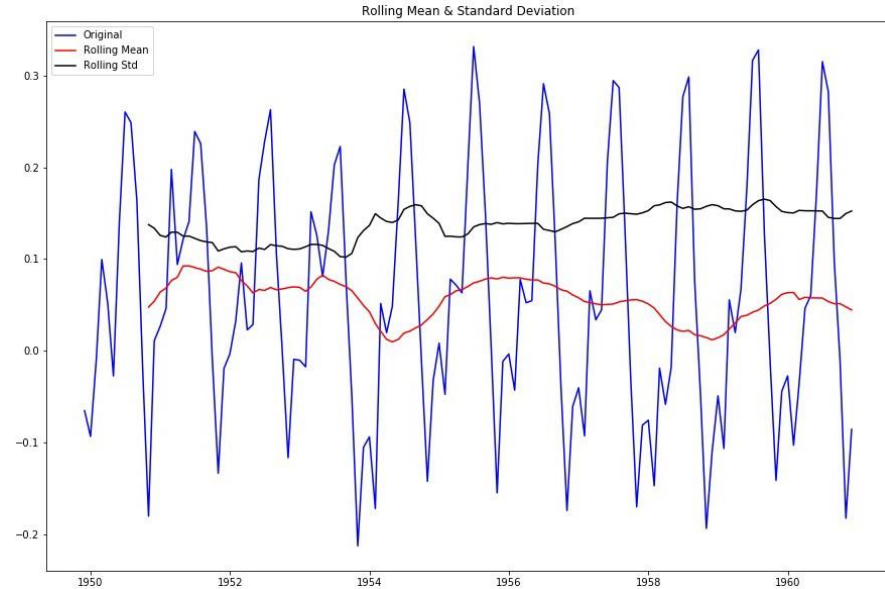
- If we remove the *trend*, we assume that the remainder of variation is due to noise and seasonality
 - It's possible that there are multiple “seasons”, we need to find the most important one (more on this later). Depending on the dataset, we might be able to make some assumptions minute, hour, day, week, etc.
- In general, we do not know if the time series is additive or multiplicative - again, we need to try them out and see what works best
 - $s_i + n_i = y_i - t_i$ or $s_i \times n_i = y_i / t_i$??
 - It's okay if we get it wrong, we can come back to this part

Airline passengers detrending

Trend is defined using a 12-month SMA



- Results of Dickey-Fuller Test:
 - Test Statistic -3.162908
 - p-value 0.022235
 - #Lags Used 13.000000
 - Number of Observations Used 119.000000
 - Critical Value (1%) -3.486535
 - Critical Value (5%) -2.886151
 - Critical Value (10%) -2.579896

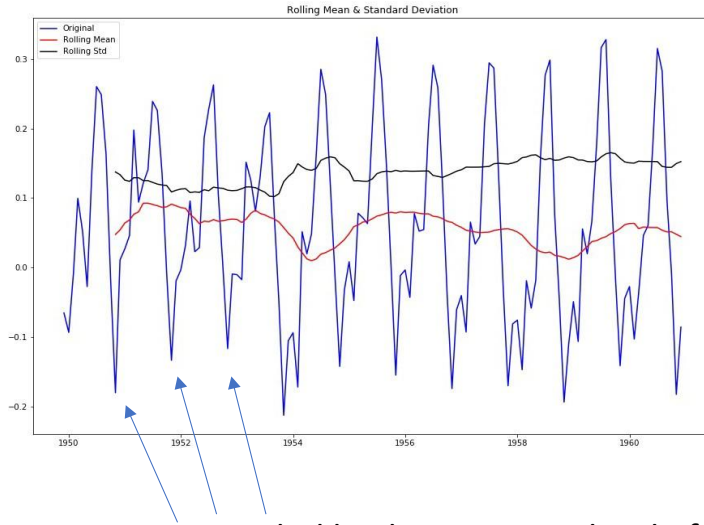


Identifying seasonality

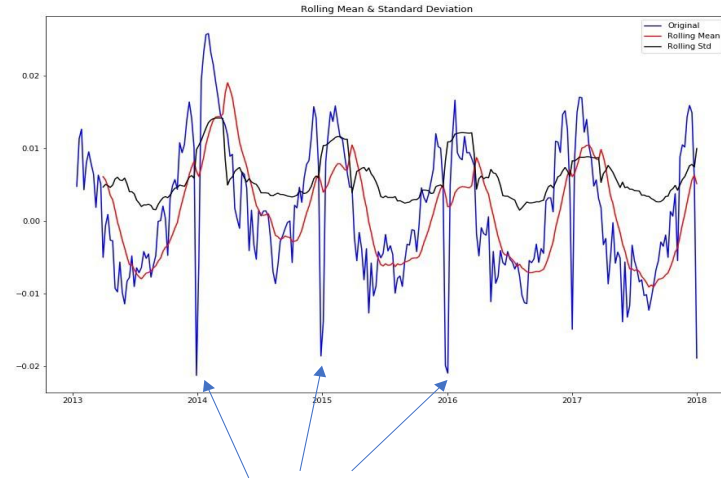
- We now have a new series containing (hopefully) only seasonality and noise components, so we then want to extract the seasonal part
- Before we carry on
 - Perform a DF test on the series to check for stationarity
 - Plot the data and see if it looks like there is seasonality
 - We can also perform autocorrelation on the data to look for stationarity (see next slide)
- The easiest way is to visually inspect the data and see if there are any clear seasonal patterns
 - We'll look at some other methods too – power spectrum
- If the data now seems to exhibit seasonality, and we can identify a reasonable period, we can use the following approach:
 - Calculate the average value for each period (eg each January, or each Monday, as applicable), and subtract this (or divide) from the series to remove the seasonality

Seasonality Examples

Airline data



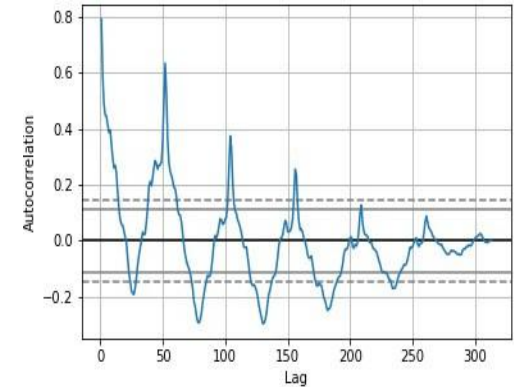
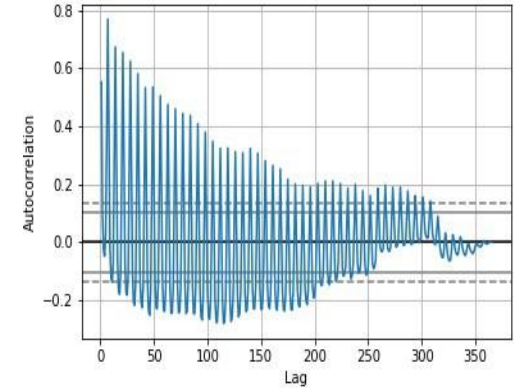
Power Consumption



Looks like there is some level of seasonality here – about 12 months

Autocorrelation

- Another way to look for seasonality is autocorrelation
- This gives the similarity between two observations as a function of the time lag between them
- The top plot is on a daily level – there is a weekly correlation
- The bottom plot is on a weekly level – there is a pattern every 54 weeks i.e year

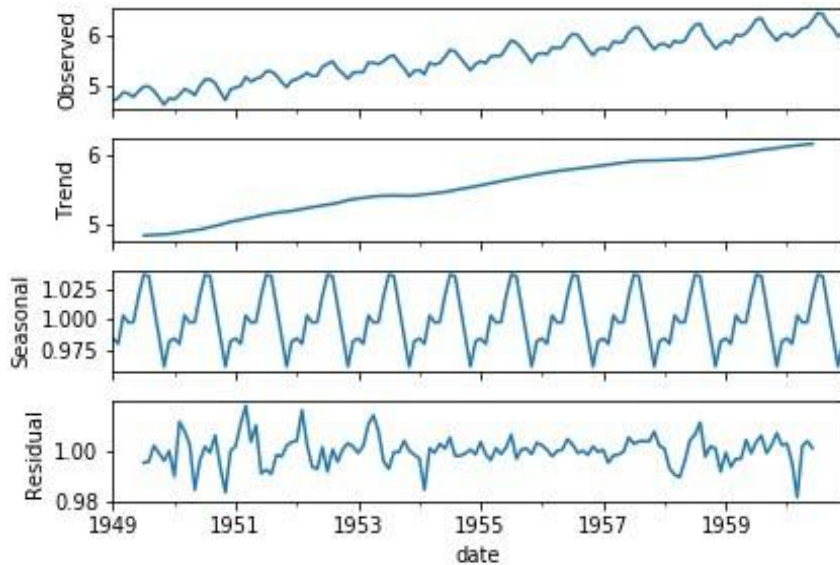


Removing seasonality

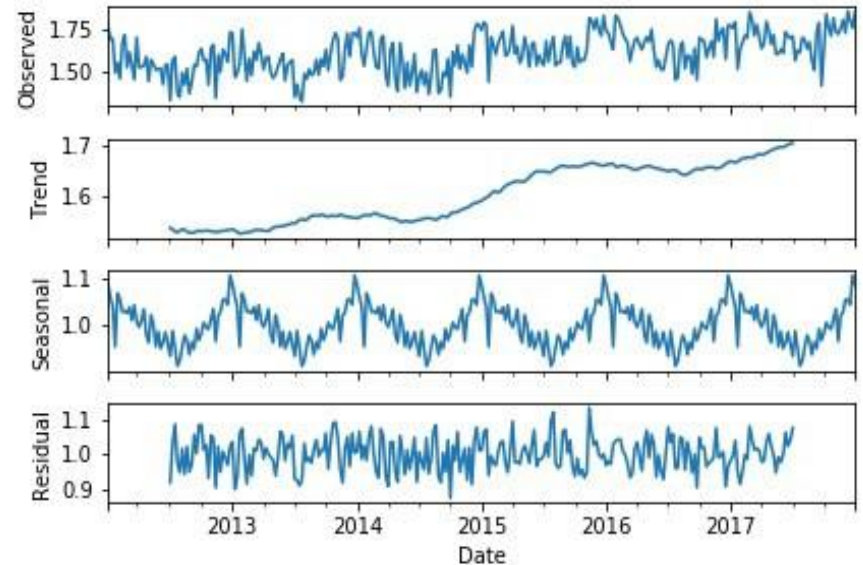
- We have now calculated what we think to be the seasonal component, so we can again remove this from our detrended dataset:
 - $n_i = y_i - t_i - s_p$ or $n_i = y_i / (t_i s_i)$
- The residual from this should give us a time series without seasonal or trend components
- There are still two checks we need to do:
 - Test for stationarity
 - Test the assumption of the additive vs multiplicative assumption: compare the addition/multiplication of each component with the original dataset
- If the residuals are stationary – awesome! We can work with this data and apply our (S)ARIMA models, or our frequency feature-based models (still to come!)

Example time decompositions

Airline Passengers



Electricity Consumption

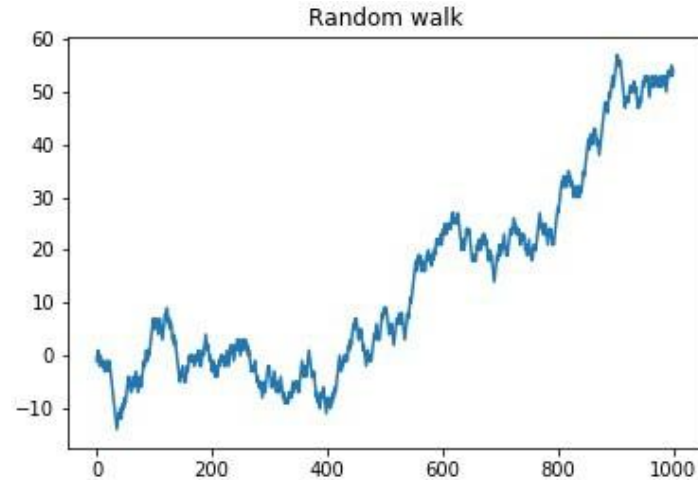


Identifying cyclical patterns

- In the first part, we assumed that cycles could be included with the trend component, and this can be sufficient
- Cycles are different from seasonal patterns, because there are no fixed periods. Alternatively, it could be the case that there are multiple “seasons” within the data, each with a different period, which gives the impression that it is not seasonal
- One way to identify these properties is by using a frequency-based approach
 - Fourier transforms, wavelets, spectral analysis

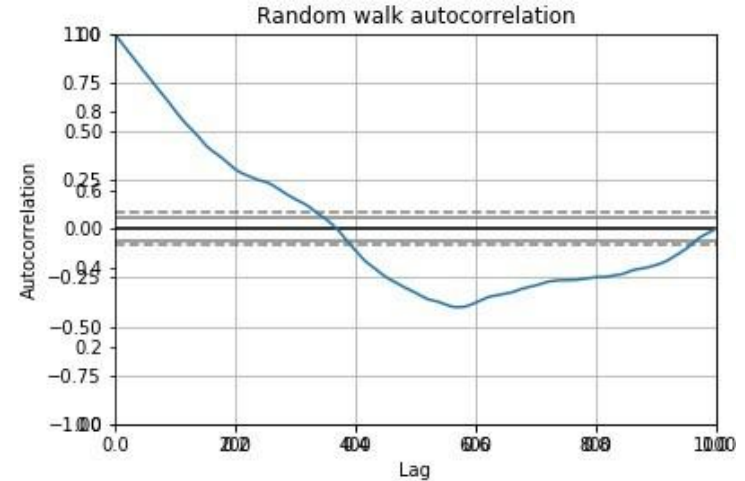
Random walks

- What if the data has no trend or seasonality? What if it really has no structure?
- A random walk is not a bunch of random numbers – each new entry has some dependence on the previous timestamp:
 - $y(t) = B_0 + B_1 * y(t - 1) + e(t)$ where B represents drift and the weighting coefficient of the previous timestamp, and e is some random fluctuation
- A random walk will be non-stationary – each step is dependent on the previous one, and therefore a function of time



Random walks

- In order to check for a random walk
 - Check the autocorrelation – if it has a strong dependence at a small lag, which decays approx. linearly, this is a sign of a random walk
 - Time series is non-stationary, and after applying techniques to make it stationary there is no obvious structure (differencing, here)
 - Using a persistence model gives better results than any other model
 - Predict every value as being equal to the previous



- Results of Dickey-Fuller Test:
 - Test Statistic -31.084824
 - p-value 0.000000
 - #Lags Used 0.000000
 - Number of Observations Used 998.000000
 - Critical Value (1%) -3.436919
 - Critical Value (5%) -2.864440
 - Critical Value (10%) -2.568314

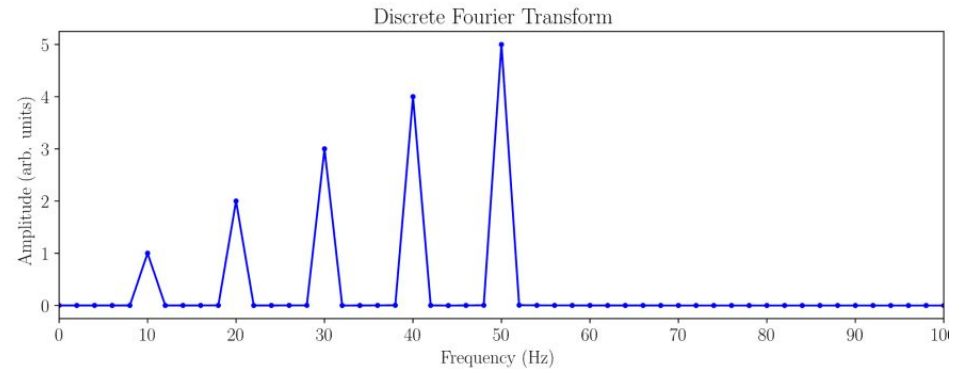
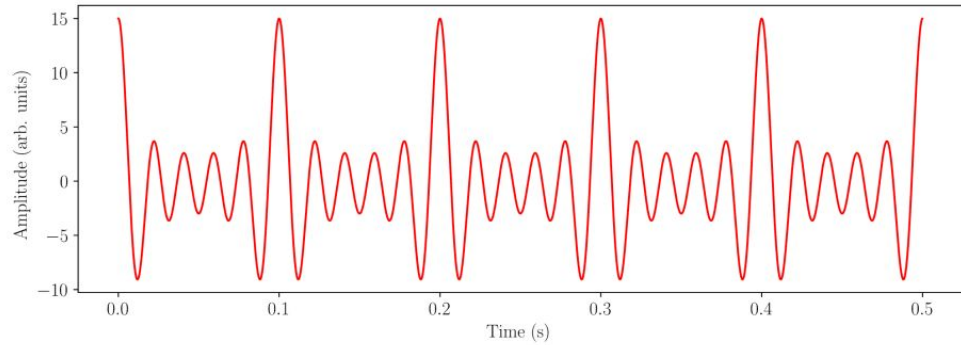
Spectral Analysis - Fourier Transforms

- <https://www.youtube.com/watch?v=spUNpyF58BY>
- Time and frequency have a fundamental relation: frequency is a measure of how often a specific event occurs over a unit of time, given an inverse relation of $f=1/T$
- If we have a time series (or any function) we can represent this in the frequency domain by using a Fourier transform
 - A FT decomposes the time series into its constituent frequencies and their relative amplitudes
 - Importantly, this assumes the time series to be stationary again
- We refer to the frequency distribution as the spectrum
- Consider a musical note: it might be played for a few seconds, but corresponds to a certain frequency
 - Similarly, when a chord is played on a guitar it consists of multiple notes from each string. The recording of the sound from the guitar is in the time domain, but if we want to know which *notes* were played, we can apply a Fourier transform to decompose this into its frequency components

Fourier Transforms

$$f(t) = F[\omega] = \left[\int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt \right]$$

$$\sum_{n=1}^5 n \cos(n\omega t), \quad \omega = 10 \times 2\pi$$

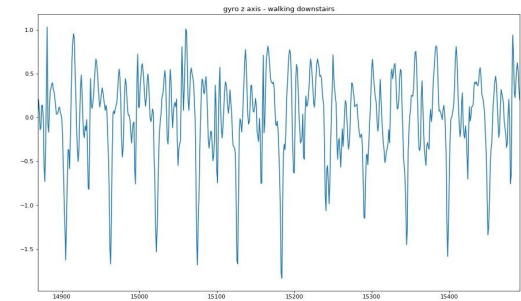
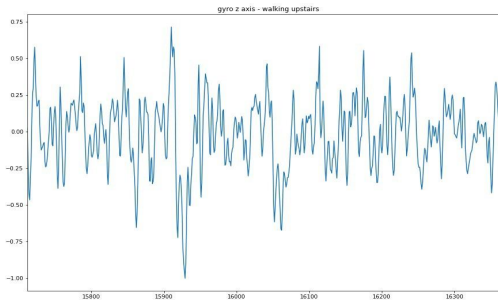
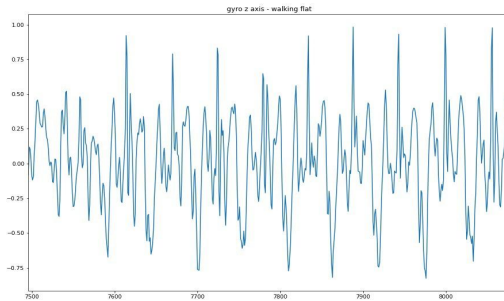


Fourier analysis in time series data

- How does this relate to what we've been looking at up until now?
- If there is some level of seasonality in the data, it will show up with a specific frequency (1 month, 1 day, etc)
- The useful thing here is that we will find **all** of the frequencies in the data, not just the main one
 - If there are multiple seasons or cycles, these will show up here
- One way in which to view this is with a spectral density plot
 - Power spectral density: this shows us how the the power of a signal is distributed by frequency, ie. which frequency components are the most informative

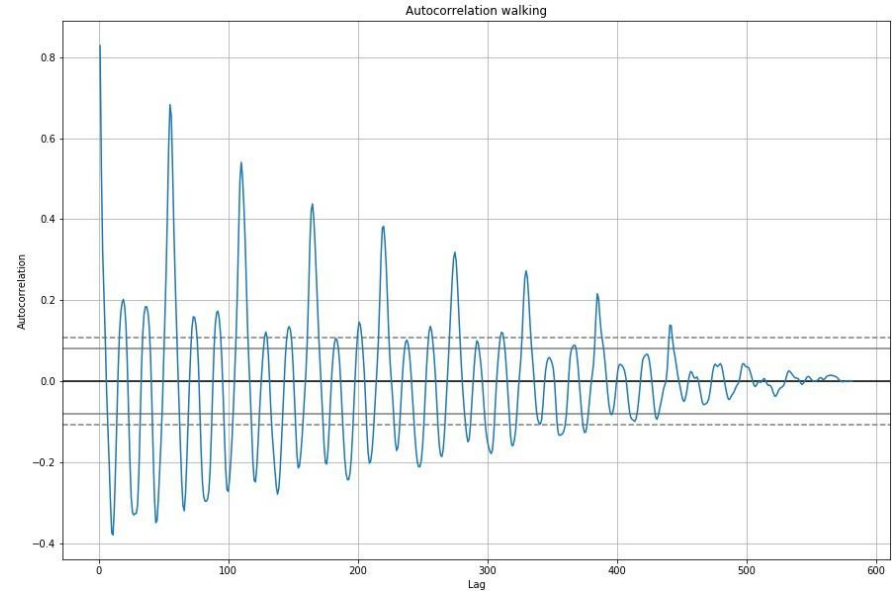
Gyro recordings for walking

- Gyro readings (z-axis only) taken from a phone at 50Hz, when a person is walking on flat ground, up & down stairs
- Expect there to be some level of seasonality – the cadence at which the person walks
 - This is clearly visible in the plots below at approx. 50 readings ~ 1Hz
- Unless they are going from a walk to a run, there shouldn't be a trend exhibited
- Interesting in the below (which makes sense), is that the cadence is lower for going up stairs, and so is the amplitude, which makes sense. For going down stairs there are peaks in the downward direction



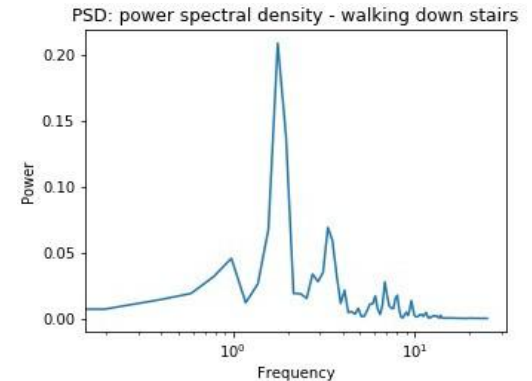
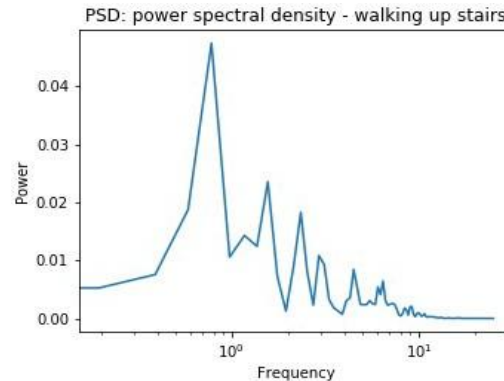
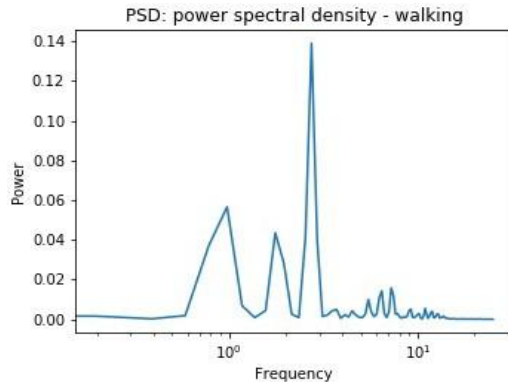
Gyro seasonality

- As before, we can look at the autocorrelation to identify the frequency at which the person takes steps – the seasonality
- Like we saw in the previous slide, there is a large correlation at about 50-60, so around 1Hz



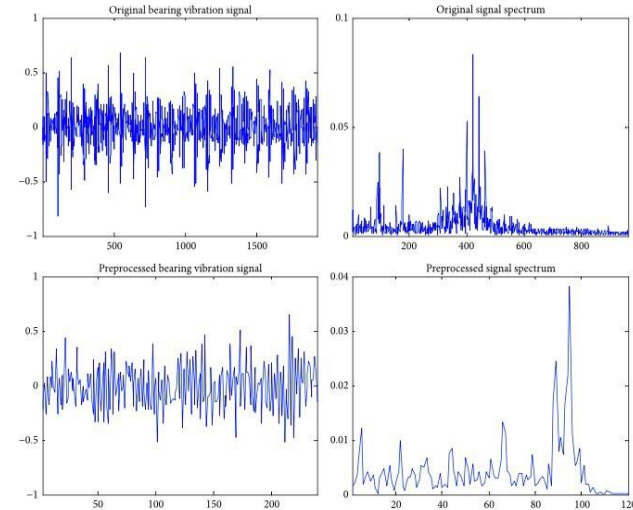
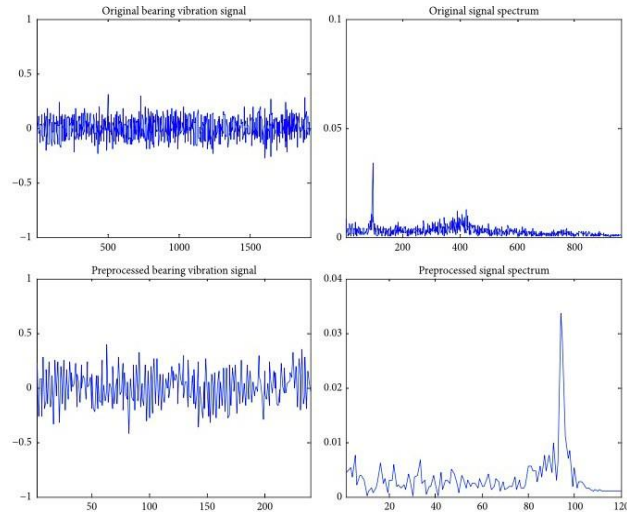
Power spectrum for walking

- Another way to view this is to look at the power spectrum
- We can see that there is a large peak in power at:
 - 1.7Hz for walking
 - 0.07Hz for walking up stairs
 - 0.9Hz for walking down stairs
- Just using the frequency we could build a model that distinguishes between all three!



Power spectrum: ball bearings

- Another example of using the frequency spectrum to find signals is looking at the vibrations coming from ball bearings
- When the ball bearing is working, there is a flat(ish) spectrum, and the frequency only spikes at around 100Hz
- When there is a problem with the bearing, there are more frequencies present, and there is obvious seasonality

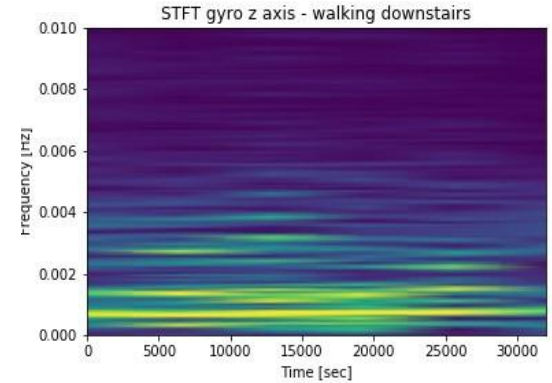
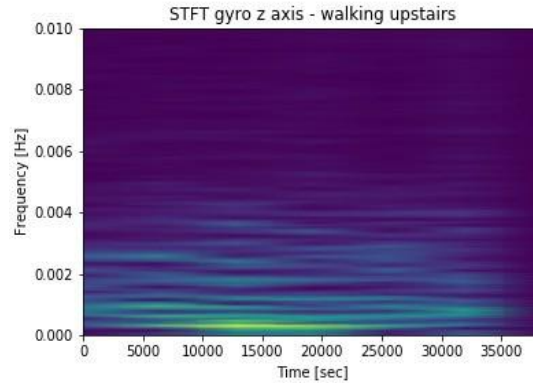
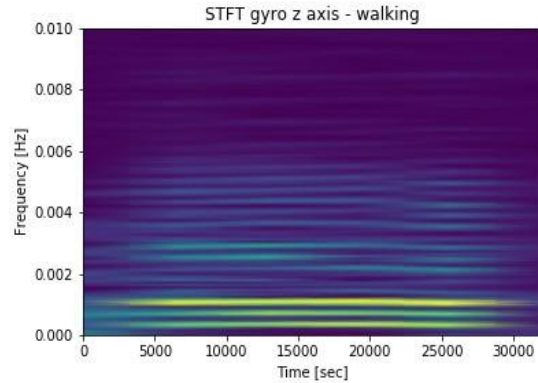


Spectrogram

- This is a visualisation technique used to show how the frequency components change over time
 - It is unlikely that many complex datasets will be stationary globally, but, if we break it up into smaller windows, each of these windows might be a better approximation of a stationary process
- Essentially, what we do is divide the time signal into a bunch of (non-) overlapping, fixed-size windows, and perform a Fourier transform on each window
 - This is called a **Short Time Fourier Transform**
 - Overlapping windows are used if there is sufficient data
- Then, each of these spectra is plotted next to each other, showing how it changes over time – this is known as a spectrogram
- It is displayed as a heatmap, with time generally on the x-axis, frequency on the y-axis, and intensity shown as a colour
- The frequency and amplitude axes can be either linear or log depending on what the graph is being used for
- Where would this be useful?
 - Something like accelerometer data, where the signal is **not** stationary, but in small enough time windows it is. It can be useful for picking up underlying signals: example could be a smart watch, which records arm movements, but, if someone is walking whilst wearing the watch, there is a likely a specific frequency in all of the windows showing the walking frequency

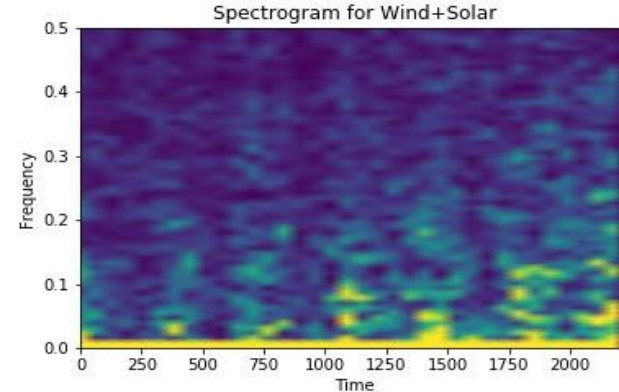
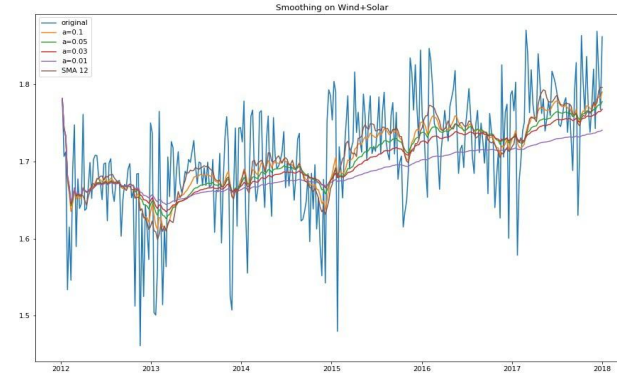
Spectrogram examples: walking

- This is similar information to the power spectrum, except it shows us how the frequency changes by time. This is extremely useful for looking at non-stationary data!
- There isn't any obvious change over time for the normal walking,
- Walking down stairs shows some deviations around the 14k mark – maybe the person started walking down the stairs too fast and had to slow down? © It gives us more insight than just the PSD though!



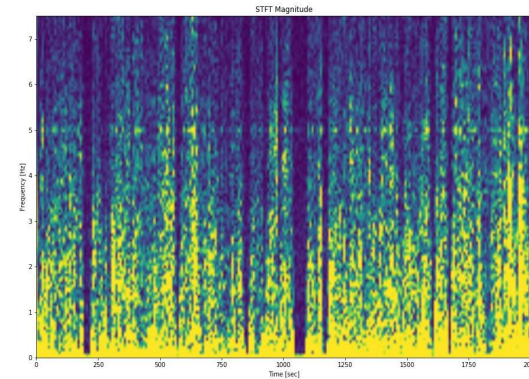
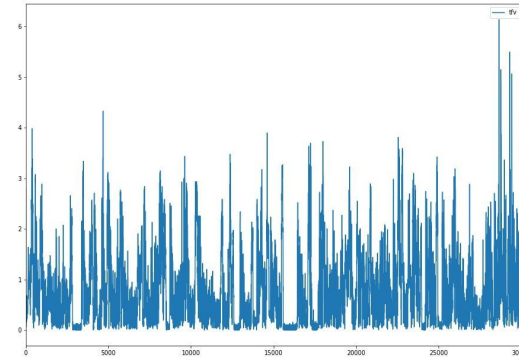
Wind & Solar Frequency

- If we look at the wind+solar graph, we know that there is some interaction between the two, and that they are not in phase with each other
- Finding the frequency/seasonality here is a bit more difficult, but there is some pattern present in the spectrogram – although it does seem to still find annual resonances



Vehicle acceleration

- Another example here is the g-force experienced by a car
- There will be spikes when accelerating or braking
- There will be some level of road noise too, but that will change based on the road type
 - We could estimate the road type by looking at the speed actually
- Interesting peak at around 5Hz though – engine noise, damaged tyre?



Next lectures

- Linear dimensionality reduction
- Wavelets
- Autoencoders
- Feature-based time series analysis